

Success Metrics for AI

◆ 1. Task Effectiveness

- **Goal Completion Rate** → % of tasks successfully completed by the workflow without human intervention.
- **Accuracy / Correctness** → How well the workflow output aligns with the ground truth, user intent, or evaluation criteria.
- **Relevance & Quality** → Human or automated scoring of response usefulness, clarity, and depth.
- **Hallucination Rate** → Frequency of factually incorrect or fabricated outputs.

◆ 2. Efficiency

- **Latency / Response Time** → Time taken to produce an answer or complete the workflow.
- **Throughput** → Number of tasks completed per unit time (important for high-volume systems).
- **Cost per Task** → Measured in API usage, compute, and memory costs.
- **Action Efficiency** → Ratio of necessary vs unnecessary tool calls (e.g., fewer redundant API calls).

◆ 3. Reliability & Robustness

- **Failure Rate** → % of workflows that terminate unexpectedly or get stuck in loops.
- **Recovery Rate** → % of failures that are automatically recovered by retries or fallback strategies.
- **Consistency** → Stability of outputs when given similar inputs (low variance in results).
- **Edge Case Handling** → How well the workflow handles incomplete, ambiguous, or noisy inputs.

◆ 4. Autonomy & Adaptability

- **Human Intervention Rate** → % of workflows requiring human help to finish.
- **Adaptability Score** → Ability to modify its plan or actions when context changes.

- **Exploration vs Exploitation Balance** → For multi-step reasoning agents, whether it over-explores or reuses known strategies effectively.

◆ 5. Collaboration (for Multi-Agent Workflows)

- **Coordination Efficiency** → How well agents communicate and minimize redundant work.
- **Conflict Resolution Rate** → How often agents resolve conflicting outputs correctly.
- **Specialization Utilization** → Degree to which agents use their domain expertise instead of duplicating efforts.

◆ 6. User & Business Impact

- **User Satisfaction (CSAT/NPS)** → Subjective rating of workflow usefulness.
- **Task Coverage** → % of user scenarios or business cases supported by the workflow.
- **Business ROI** → Reduction in cost, increase in revenue, or efficiency gained from automation.
- **Trustworthiness** → User confidence that outputs are reliable and safe.

◆ 7. Safety & Compliance

- **Toxicity Rate** → % of outputs flagged as unsafe, biased, or non-compliant.
- **Data Privacy Adherence** → No leakage of sensitive information across steps.
- **Policy Violation Rate** → Alignment with company or regulatory policies.